Heterogeneous Ensemble Learning for AI-Based Classification of NIDS Alerts*

Dohyun Ahn, Chungil Cha, and Hayoung Seong

Korea Institute of Science and Technology Information (KISTI), Daejeon, Republic of Korea
{dksehgus0122, cicha, shy2028}@kisti.re.kr

Abstract

In security operation centers (SOCs), high rates of false positives in Network Intrusion Detection System (NIDS) alerts increase analysts' workload and reduce monitoring efficiency. However, many existing approaches rely on either statistical features or raw payload data alone, limiting their ability to exploit heterogeneous NIDS log information. This study proposes a feature-aware heterogeneous ensemble method for true-positive and false-positive (TP/FP) classification using real-world KISTI NIDS logs. The method combines packet metadata (PM), payload byte frequency (PBF), and raw payload (RP). ML models process structured features, while DL models learn raw payload patterns. The 1D-CNN + XGBoost ensemble using PM + PBF + RP achieved the best performance, with accuracy of 0.9818, an F1-score of 0.9782, and an AUC of 0.9949. These results show that heterogeneous feature integration improves AI-based NIDS alert classification.

*Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-37, July 8-10, 2026, Seoul, South Korea.
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